

Aayush Ahuja

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Summary

Staff software engineer (IC6) with 10+ years building large-scale detection, ML, and agentic systems. Deep experience across distributed data infrastructure, ML/LLM infrastructure, and safety-critical systems, with a strong track record of technical leadership across large, cross-functional orgs. Currently leading the adoption of autonomous multi-agent systems for large-scale system monitoring, investigation, and remediation across multiple teams at Meta.

Experience

Software Engineer, IC6 — Integrity — Meta

Feb 2019 – Present

Technical direction across large-scale systems, ML infrastructure, and autonomous agent platforms; technical scope spans 30+ engineers

- Leading design and delivery of a cross-org autonomous multi-agent self-learning system for large-scale issue discovery, adopted across 13+ teams: agents search and discover problems in production systems, fan out across related surfaces, perform root-cause analysis, and propose long-term fixes; also used to respond to the majority of high-profile escalations from governments and press. Implemented multiple novel recursive self-improvement techniques and memory to make the agent more effective.
- One of the first engineers on Meta's personalized-agent safety initiative (Project Hatch, MSL); designed and shipped the initial safety/detection layer — prompt-injection defense, high-risk-harm checks, LLM-based review, human-in-the-loop escalation — during the project's ~1.5-month bootstrap phase, before returning to core integrity work
- Built and led multiple LLM-based scam-ad detection launches, including the inference infrastructure powering them
- TL of Ads Integrity domain — led cross-org collaborations on shared detection systems and signal-sharing across the payment fraud domain, driving a >\$200M/year reduction in payment fraud and a cumulative ~15-20% reduction in scam reports across initiatives
- Redesigned ML infrastructure (async processing and feature availability): cut error rates from 3-4% to 0.01% and increased enforcement recall by 1000%
- Expanded integrity engineering frameworks and skills — adopted by 1,000+ engineers; led cross-org sessions on AI-assisted engineering productivity for 200+ engineers, and trained 150+ engineers to be AI-native as part of an org-wide AI4Productivity initiative

Software Engineer II, Data Platform & Infrastructure — Uber

Jun 2018 – Jan 2019

- Scaled a Presto/Hive query gateway to 5,000+ weekly active users and 100k+ daily queries; built rate limiting, load balancing, and phased-rollout tooling for Hive deployments

Software Engineer, Spam Detection — Media.net (Directi)

Aug 2015 – Jun 2018

- Designed and built data pipelines and statistical/ML models (high-frequency, affiliation, co-visitation, crowd-sourcing, clustering) to detect spam and fraud across a large ad network processing 50M+ data points daily; led and mentored a team of 3 engineers

Member of Technical Staff, EDW — Groupon

Jul 2014 – Aug 2015

- Built ETL pipeline tooling for a Teradata/HDFS enterprise data warehouse

Education

B.E. (Hons), Computer Science — BITS-Pilani, Pilani Campus | 2010 – 2014

Skills

Python · LLM / agentic system design · Distributed systems & data infrastructure · ML infrastructure & evaluation · Large-scale detection systems · Cross-org technical leadership · AI safety & red-teaming